Namal University, Mianwali

Department of Computer Science

CSC-271 – Database Systems
Complex Computing Problem – Milestone 2

Entity-Relationship Diagram (ERD)

for

Electric Bus Location Tracker System

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Introduction

The Electric Bus Location Tracker System is a mobile-based application developed for the electric bus transportation network in Mianwali, Punjab, Pakistan. The system addresses the growing need for real-time bus tracking and centralized management of transportation operations in the region.

The system serves three primary user types: **Passengers**, **Drivers**, and **Administrators**. However, it is critical to understand how each interacts with the database:

- **Passengers** are *unauthenticated public users* of the mobile application. They can view real-time bus locations, search routes, calculate estimated arrival times, and manage favourite routes. Passengers do **not** register or log in to the system (SRS FR-005); therefore, **no passenger data is persisted in the relational database**. Favourite routes selected by passengers are stored **locally on the passenger's mobile device**, not in the database.
- **Drivers** are authenticated users who log in to manage duty schedules and share live GPS location. They are registered in the database by Administrators.
- **Administrators** are authenticated users who oversee fleet operations, assign duties, manage routes, and generate operational reports. They are registered by the Punjab Mass Transit Authority (PMA).

A well-structured relational database is the core component of this system. It must efficiently store and retrieve data related to authenticated users (Drivers and Admins), buses, routes, stops, schedules, duty assignments, and operational reports. This milestone presents the complete Entity-Relationship Diagram (ERD) for the system database, designed using **Crow's Foot notation**, along with detailed entity attributes, relationship cardinalities, and governing business rules. The ERD serves as the blueprint from which the MySQL database implementation will be derived in subsequent milestones.

Problem Statement

Passengers using the electric bus service in Mianwali currently have no reliable way to check real-time bus locations, estimated arrival times, or schedule information. This lack of information leads to wasted waiting time and general inconvenience for the travelling public.

From an operational perspective, the transportation authority (PMA) faces challenges in managing bus assignments, monitoring driver duties, tracking vehicle maintenance history, and generating operational reports — all due to the absence of a centralized data management system.

Therefore, the core problem addressed in this project is the **absence of a structured, normalized relational database** that can efficiently store, relate, and retrieve all data required to support real-time bus tracking, route and schedule management, driver duty scheduling, and administrative reporting for the Electric Bus Location Tracker System. Since only Drivers and Administrators are authenticated users (SRS FR-005), the database is scoped accordingly — passenger-facing features rely on publicly accessible data without requiring passenger records in the database.

Objectives

General Objective

To design a comprehensive, normalized relational database for the Electric Bus Location Tracker System that supports all core functional requirements defined in the SRS document.

Specific Objectives

1. **Identify and define all entities** present in the system including User, Driver, Admin, Bus, Route, Stop, Schedule, Duty_Assignment, Route_Stop_Detail, and Report.
2. **Define attributes** for each entity with appropriate data types, constraints (PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE), and default values to ensure data integrity.
3. **Model supertype–subtype relationships** by using the `User` entity as a supertype for `Driver` and `Admin` subtypes (only authenticated roles), eliminating data redundancy.
4. **Determine all relationships** between entities and specify correct cardinalities (one-to-one, one-to-many, many-to-many) using Crow's Foot notation standards.
5. **Resolve the many-to-many** Route–Stop relationship through an explicit **associative entity** (`Route_Stop_Detail`).
6. **Define business rules** to enforce data integrity, prevent scheduling conflicts, and govern status transitions within the system.
7. **Verify normalization** of the database design up to Third Normal Form (3NF) to eliminate redundancy and transitive dependencies.
8. **Map all entities and relationships** to the functional requirements defined in the SRS document to ensure complete requirement coverage.

Entities Identification

Entities are real-world objects that have a distinct existence within the context of the Electric Bus Location Tracker System's **relational database**. Since Passengers are unauthenticated and their data is not persisted (SRS FR-005), no Passenger entity exists in the database. Passenger-facing features (real-time tracking, route search, EAT) are served by reading publicly available data from existing entities (Route, Stop, Schedule, Bus) without any passenger record. Based on the SRS document and the system ERD, the following entities have been identified:

Table 1: List of Entities in the System

#	Entity Name	Description
1	User	Stores all user account information common to both Drivers and Administrators. Acts as the supertype entity in the supertype–subtype hierarchy. Only two roles exist: `Driver` and `Admin` (SRS FR-005).
2	Driver	Stores driver-specific information including license number, hire date, and duty status. Subtype of User.
3	Admin	Stores administrator-specific information for system management. Subtype of User.
4	Bus	Stores electric bus information including bus number, manufacture year, maintenance records, and operational status.
5	Route	Stores bus route definitions including starting point, destination, total distance, and estimated duration.
6	Stop	Stores bus stop locations where passengers can board or alight.
7	Route_Stop_Detail	**Associative entity** that resolves the many-to-many relationship between Route and Stop. Represents the association of stops along a route.
8	Schedule	Stores planned trip schedules linking a Route to a Bus with departure and arrival times on a given date.
9	Duty_Assignment	Stores driver duty assignments linking Drivers to Buses and Schedules, created by an Administrator.
10	Report	Stores operational reports generated or recorded by Administrators.

Supertype–Subtype Structure

The **User** entity acts as a **supertype** for the **Driver** and **Admin** subtypes. This design decision is justified by the following:

- Driver and Admin share common attributes: `User_Id`, `Name`, `Email`, `Contact`, `Password`, `Role`, `Account_Status`, `Creation_Date`, and `Deletion_Date`.
- Each subtype has distinct attributes:
  - **Driver**: `License_No`, `Hire_Date`, `Status`
  - **Admin**: no additional domain attributes beyond the FK back to User
- This structure eliminates data redundancy and satisfies normalization principles.
- The generalization is **disjoint** (a User is either a Driver or an Admin, not both) and **total** for authenticated users (every authenticated User record must correspond to exactly one subtype record).
- **Passengers are excluded** from this hierarchy entirely: they are unauthenticated and have no record in the database (SRS FR-005).

The relationship between User and each subtype is **one-to-one (1:1)**, as each User record corresponds to exactly one Driver or exactly one Admin record.

Attributes per Entity

Entity: User

Table 2: User Entity Attributes

Attribute	**Data Type**	**Constraints**	**Description**
User_Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user identifier
Name	VARCHAR(100)	NOT NULL	Full name of user
Email	VARCHAR(100)	UNIQUE, NOT NULL	Email address
Contact	VARCHAR(15)	NOT NULL	Phone number
Password	VARCHAR(255)	NOT NULL	Bcrypt-hashed password
Role	ENUM	('Driver', 'Admin'), NOT NULL	User role / subtype discriminator. Passenger role does not exist in the database (SRS FR-005).
Account_Status	ENUM	('Active', 'Inactive'), DEFAULT 'Active'	Account status
Creation_Date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp
Deletion_Date	DATETIME	NULL	Soft-deletion timestamp

Entity: Driver

Table 3: Driver Entity Attributes

Attribute	**Data Type**	**Constraints**	**Description**
Driver_Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique driver identifier
License_No	VARCHAR(50)	UNIQUE, NOT NULL	Driving licence number
Hire_Date	DATE	NOT NULL	Date of hiring
Status	ENUM	('Available', 'On-Duty', 'Off-Duty'), DEFAULT 'Available'	Current driver status
User_Id	INT	FOREIGN KEY REFERENCES User(User_Id), NOT NULL	Links to supertype User record

Entity: Admin

Table 4: Admin Entity Attributes

Attribute	**Data Type**	**Constraints**	**Description**
Admin_Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique admin identifier
User_Id	INT	FOREIGN KEY REFERENCES User(User_Id), NOT NULL	Links to supertype User record

Entity: Bus

Table 5: Bus Entity Attributes

Attribute	**Data Type**	**Constraints**	**Description**
Bus_Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique bus identifier
Bus_Number	VARCHAR(20)	UNIQUE, NOT NULL	Bus registration number
Status	ENUM	('Active', 'Maintenance', 'Inactive'), DEFAULT 'Active'	Operational status
Registration_Date	DATE	NOT NULL	Date of registring bus to Punjab Masstransit Authority, Mianwali

Entity: Route

Table 6: Route Entity Attributes

Attribute	**Data Type**	**Constraints**	**Description**
Route_Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique route identifier
Name	VARCHAR(100)	NOT NULL	Route name
Starting_Point	VARCHAR(100)	NOT NULL	Origin location
Destination_Point	VARCHAR(100)	NOT NULL	End location
Distance	DECIMAL(10,2)	NOT NULL	Total distance in km
Estimated_Duration	INT	NOT NULL	Duration in minutes
Status	ENUM	('Active', 'Inactive'), DEFAULT 'Active'	Route status

Entity: Stop

Table 7: Stop Entity Attributes

Attribute	**Data Type**	**Constraints**	**Description**
Stop_Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique stop identifier
Name	VARCHAR(100)	NOT NULL	Stop name
Creation_Date	DATETIME	DEFAULT CURRENT_TIMESTAMP	When stop was added
Deletion_Date	DATETIME	NULL	Soft-deletion timestamp
Status	ENUM	('Active', 'Inactive'), DEFAULT 'Active'	Stop status

Associative Entity: Route_Stop_Detail

The many-to-many (M:N) relationship between `Route` and `Stop` is resolved through the **Route_Stop_Detail associative entity**. In Crow's Foot notation, an associative entity is represented as a table that sits between two entities and holds the foreign keys of both as a composite primary key. This is explicitly shown in the ERD.

Table 8: Route_Stop_Detail Associative Entity Attributes

Attribute	**Data Type**	**Constraints**	**Description**
Route_Id	INT	PRIMARY KEY (composite), FOREIGN KEY REFERENCES Route(Route_Id), NOT NULL	Associated route
Stop_Id	INT	PRIMARY KEY (composite), FOREIGN KEY REFERENCES Stop(Stop_Id), NOT NULL	Associated stop

Entity: Schedule

Table 9: Schedule Entity Attributes

Attribute	**Data Type**	**Constraints**	**Description**
Schedule_Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique schedule identifier
Route_Id	INT	FOREIGN KEY REFERENCES Route(Route_Id)	Associated route
Bus_Id	INT	FOREIGN KEY REFERENCES Bus(Bus_Id)	Assigned bus
Departure_Time	TIME	NOT NULL	Scheduled departure time
Arrival_Time	TIME	NOT NULL	Scheduled arrival time
Date	DATE	NOT NULL	Date of the scheduled trip

Entity: Duty_Assignment

Table 10: Duty_Assignment Entity Attributes

Attribute	**Data Type**	**Constraints**	**Description**
Duty_Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique duty identifier
Driver_Id	INT	FOREIGN KEY REFERENCES Driver(Driver_Id), NOT NULL	Assigned driver
Bus_Id	INT	FOREIGN KEY REFERENCES Bus(Bus_Id), NOT NULL	Assigned bus
Schedule_Id	INT	FOREIGN KEY REFERENCES Schedule(Schedule_Id), NOT NULL	Associated schedule
Start_Time	TIME	NOT NULL	Actual duty start time
End_Time	TIME	NULL	Actual duty end time
Status	ENUM	('Scheduled', 'In-Progress', 'Completed'), DEFAULT 'Scheduled'	Duty status
Admin_Id	INT	FOREIGN KEY REFERENCES Admin(Admin_Id), NOT NULL	Admin who created duty

Entity: Report

The `Report` table holds the foreign key `Admin_Id`, which correctly models the one-to-many relationship between Admin and Report (one Admin can generate many Reports).

Table 11: Report Entity Attributes

Attribute	**Data Type**	**Constraints**	**Description**
Report_Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique report identifier
Admin_Id	INT	FOREIGN KEY REFERENCES Admin(Admin_Id), NOT NULL	Admin who generated report
Type	VARCHAR(50)	NOT NULL	Report type (daily/weekly/monthly)
Time_And_Date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Date and time of generation
Description	TEXT	NULL	Report content or summary

Relationships and Cardinalities

Overview

In **Crow's Foot notation**, relationship cardinalities are expressed using the following symbols on the line connecting two entities:

- **One** side: a single vertical bar (|) denotes "exactly one"
- **Many** side: a crow's foot (three-pronged fork) denotes "many"
- **Mandatory** participation: a single vertical bar close to the entity
- **Optional** participation: a small circle (zero) close to the entity

All relationships in the ERD follow these conventions. Relationship names used in the table below match the labels displayed on the ERD diagram exactly.

Relationship Summary Table

Table 12: Complete Relationship List with Cardinalities (Crow's Foot Notation)

Relationship Name	**From Entity**	**To Entity**	**Type**	**Min : Max**
Is_A	User	Driver	One-to-One	0,1 : 1,1
Is_A	User	Admin	One-to-One	0,1 : 1,1
Performs	Driver	Duty_Assignment	One-to-Many	1,1 : 0,N
Assigns	Admin	Duty_Assignment	One-to-Many	1,1 : 0,N
Assigned_To	Bus	Duty_Assignment	One-to-Many	1,1 : 0,N
has	Schedule	Duty_Assignment	One-to-Many	1,1 : 0,N
Assigned_To	Bus	Schedule	One-to-Many	0,1 : 0,N
Added_To	Route	Schedule	One-to-Many	1,1 : 0,N
has	Route	Route_Stop_Detail	One-to-Many	1,1 : 1,N
Included_In	Stop	Route_Stop_Detail	One-to-Many	1,1 : 0,N
generates	Admin	Report	One-to-Many	1,1 : 0,N

Detailed Relationship Descriptions

User – Driver (*Is_A*, One-to-One)

Each User can optionally be associated with at most one Driver record (minimum 0, maximum 1 on the Driver side). Each Driver must be associated with exactly one User record (minimum 1, maximum 1 on the User side). The ERD labels this relationship **Is_A**. This enforces the supertype–subtype integrity: the User table stores the common authentication data and the Driver table stores role-specific data.

User – Admin (*Is_A*, One-to-One)

Each User can optionally be associated with at most one Admin record. Each Admin must be associated with exactly one User record. The ERD labels this relationship **Is_A**. Same Crow's Foot pattern as the User–Driver relationship above.

Driver – Duty_Assignment (*Performs*, One-to-Many)

A Driver can have zero or many Duty Assignments over time (minimum 0, maximum N on the Duty_Assignment side). Each Duty Assignment must be assigned to exactly one

Driver (minimum 1, maximum 1 on the Driver side). The ERD labels this relationship **Performs**: a driver *performs* duty assignments.

Admin – Duty_Assignment (*Assigns*, One-to-Many)

An Administrator can create zero or many Duty Assignments. Each Duty Assignment is created by exactly one Administrator. The ERD labels this relationship **Assigns**. This enforces business rule BR-06: only admins can create duty records (SRS FR-067, FR-068, FR-069).

Bus – Duty_Assignment (*Assigned_To*, One-to-Many)

A Bus can appear in zero or many Duty Assignments across different dates. Each Duty Assignment references exactly one Bus. The ERD labels this relationship **Assigned_To**: a duty assignment is *assigned to* a bus.

Schedule – Duty_Assignment (*has*, One-to-Many)

A Schedule can be referenced by zero or many Duty Assignments. Each Duty Assignment is linked to exactly one Schedule. The ERD labels this relationship **has**.

Bus – Schedule (*Assigned_To*, One-to-Many)

A Bus can be scheduled for zero or many trips over time. Each Schedule entry references exactly one Bus. The ERD labels this relationship **Assigned_To**: a schedule is *assigned to* a bus.

Route – Schedule (*Added_To*, One-to-Many)

Many Schedules can be added to the same Route (a route can be served at multiple times or on multiple dates). Each Schedule belongs to exactly one Route. The ERD labels this relationship **Added_To**: a schedule is *added to* a route.

Route – Route_Stop_Detail (*has*, One-to-Many)

A Route must contain at least one stop association (minimum 1, maximum N on the associative entity side). Each Route_Stop_Detail record belongs to exactly one Route. The ERD labels this relationship **has**.

Route_Stop_Detail – Stop (*Included_In*, Many-to-One)

Each Route_Stop_Detail record references exactly one Stop. A Stop can be referenced in zero or many Route_Stop_Detail records (a stop may exist independently before being assigned to any route). The ERD labels this relationship **Included_In**: a stop is *included in* a route's stop list.

Admin – Report (*generates*, One-to-Many)

An Administrator can generate zero or many Reports. Each Report must be linked to exactly one Administrator. The ERD labels this relationship **generates**. The foreign key (`Admin_Id`) resides in the `Report` table — **not** in the `Admin` table — because this is a one-to-many relationship. This design correctly supports SRS FR-072 through FR-075.

Resolved Many-to-Many: Route and Stop

In Crow's Foot notation, a many-to-many (M:N) relationship between two entities is **always resolved** into two one-to-many relationships through an **associative entity**. The Route–Stop M:N is resolved as follows:

$$\textbf{Route} \xrightarrow{\textit{has}\ (1{:}N)} \textbf{Route_Stop_Detail} \xrightarrow{\textit{Included_In}\ (N{:}1)} \textbf{Stop}$$

The `Route_Stop_Detail` associative entity holds the composite primary key `(Route_Id, Stop_Id)`, uniquely identifying each Route–Stop pair. This structure is explicitly shown in the ERD.

Entity-Relationship Diagram (ERD)

The ERD was created using **Crow's Foot notation**, which is the industry-standard notation for relational database design. The following symbols are used:

Table 13: Crow's Foot Notation Legend

Symbol / Convention	**Meaning**
Rectangle (table box)	Entity
Solid underlined attribute	Primary Key
Dotted underline attribute	Foreign Key
`||---<o`	One-to-Many: mandatory one, optional many
`o|---<o`	One-to-Many: optional one, optional many
`o|---|o`	One-to-One: both sides optional
Rectangle with PK+FK columns	Associative entity (resolves M:N)

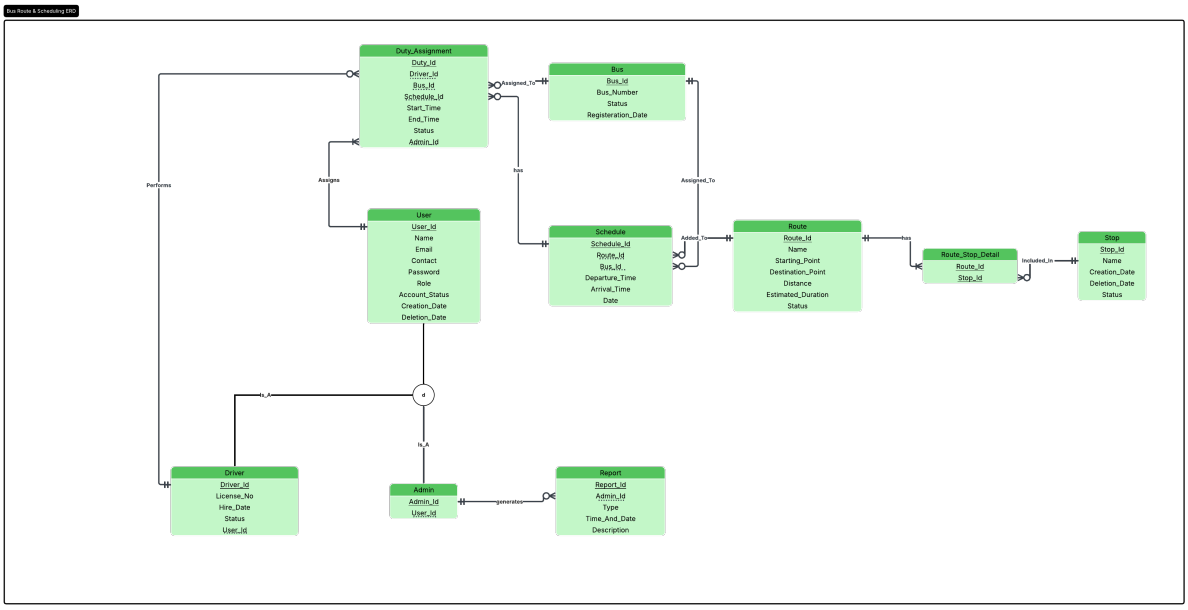

Figure 1: Complete ERD for Electric Bus Location Tracker System (Crow's Foot Notation)

ERD Component Summary

The following counts reflect the **correct intended design** (with the two ERD corrections applied):

Table 14: ERD Statistics (Intended Correct Design)

Component	**Count**
Regular Entities	9
Associative Entities	1
Total Entities (incl. associative)	10
Relationships	11
Total Attributes	52
Primary Keys	10
Foreign Keys	11

Business Rules

The following business rules govern the database design and enforce data integrity across all entities and relationships:

BR-1. User Role Separation: A User can be either a Driver or an Admin, not both simultaneously. The `Role` ENUM (`'Driver'`, `'Admin'`) acts as the subtype discriminator, enforced through the *Is_A* one-to-one relationships. Passengers are not represented as users in the database (SRS FR-005).

BR-2. Driver Duty Conflict Prevention: A Driver cannot have two overlapping Duty Assignments at the same time on the same date. The system must validate time overlaps before inserting a new Duty Assignment record (*Performs* relationship; SRS FR-070).

BR-3. Bus Availability Constraint: A Bus cannot be assigned to multiple concurrent duties that overlap in time. Each Bus can only be at one location at any given time (*Assigned_To* relationship; SRS FR-071).

BR-4. Soft Deletion Policy: Setting `Deletion_Date` in the `User` or `Stop` table marks the record as logically deleted while preserving it for audit and historical purposes. Active records have a `NULL` Deletion_Date (SRS FR-066).

BR-5. Duty Status Flow: Duty status must follow a strict progression: `Scheduled` → `In-Progress` → `Completed`. No backward transitions are permitted (SRS FR-039, FR-046).

BR-6. Admin-Only Duty Creation: Only an Administrator can create or modify Duty Assignment records (*Assigns* relationship; SRS FR-067, FR-068, FR-069). Drivers have no write access to this entity.

BR-7. Route–Stop Minimum: A Route must be associated with at least one Stop via the Route_Stop_Detail associative entity (*has* and *Included_In* relationships). A Stop may exist independently before being assigned to any Route.

BR-8. Driver Licence Uniqueness: Each Driver must have a unique `License_No`. The UNIQUE constraint on this attribute prevents duplicate licence entries.

BR-9. Bus Number Uniqueness: Each Bus must have a unique `Bus_Number` across the entire system. The UNIQUE constraint enforces this.

BR-10. Driver Status Automation: When a Driver starts a duty, their `Status` automatically transitions from `Available` to `On-Duty`. Upon duty completion, status reverts to `Available` (*Performs* relationship; SRS FR-042, FR-047).

BR-11. Driver Account Deletion Restriction: A **Driver** account cannot be physically deleted from the database if it has active Duty Assignment records linked through the *Performs* relationship (SRS FR-066). Only deactivation (setting `Account_Status = 'Inactive'`) is permitted in such cases. This rule applies to Driver accounts managed by Administrators under User Account Management (SRS Section 3.4.3).

BR-12. Report Ownership via *generates*: Every Report record must be associated with a valid `Admin_Id` via the foreign key in the `Report` table. Reports cannot exist without an owning Administrator.

BR-13. Schedule–Bus–Route Active Status: A Schedule entry (linked via *Assigned_To* to Bus and *Added_To* to Route) must reference a Bus and a Route that are both currently in `Active` status. Inactive buses or routes cannot be scheduled for new trips.

BR-14. No Passenger Data in Database: Passengers are unauthenticated public users (SRS FR-005). The system serves passenger-facing features (real-time bus tracking, route search, EAT calculation) by exposing read-only data from the `Route`, `Stop`, `Schedule`, and `Bus` entities. No passenger records, sessions, or profiles are stored in the relational database.

BR-15. Favourite Routes Stored Locally: The Manage Favourite Routes feature (SRS FR-029 to FR-032) allows passengers to save and remove favourite routes. Since passengers are not authenticated and have no database record, favourite route data is persisted **locally on the passenger's mobile device** (e.g., in device storage). This data is **not** stored in the server-side relational database.

Mapping to System Requirements

Table 15: Requirements to ERD Mapping

Requirement Category	**Implemented Via**
User Authentication & Authorization (FR-001–FR-008)	User table (Email, Password, Role, Account_Status) supporting JWT-based authentication for Driver and Admin roles only via *Is_A* relationships. Passengers are not authenticated (SRS FR-005).
Driver Management (FR-033–FR-048)	Driver table linked to User via *Is_A* (License_No, Hire_Date, Status); Duty_Assignment table via *Performs* for duty scheduling and tracking.
Admin Management	Admin table linked to User via *Is_A*.
Bus Management (FR-049–FR-054)	Bus table (Bus_Number, Manufacture_Year, Status, Last_Maintainence_Date).
Route Management (FR-055–FR-060)	Route table (Starting_Point, Destination_Point, Distance, Estimated_Duration); Schedule table linked via *Added_To*.
Stop Management (FR-056, FR-058)	Stop table with soft-deletion support (Deletion_Date, Status); linked to routes via *Included_In*.
Schedule Management (FR-059–FR-060)	Schedule table linking Route (*Added_To*) and Bus (*Assigned_To*) with Departure_Time, Arrival_Time, Date.
Duty Assignment (FR-067–FR-071)	Duty_Assignment table linking Driver (*Performs*), Bus (*Assigned_To*), Schedule (*has*), and Admin (*Assigns*); conflict prevention enforced by BR-02 and BR-03.
Operational Reports (FR-072–FR-075)	Report table linked to Admin via *generates* relationship (Type, Time_And_Date, Description).
Route–Stop Association (FR-056, FR-018)	Route_Stop_Detail associative entity resolving the M:N relationship via *has* and *Included_In*.
Passenger Bus Tracking (FR-010–FR-013)	Served by reading Bus table (status, location) and Schedule table. No passenger entity required.
Passenger Route & Schedule Search (FR-014–FR-022)	Served by reading Route, Stop, Schedule, and Route_Stop_Detail tables. No passenger entity required.
Passenger EAT Calculation (FR-023–FR-028)	Computed from Schedule (*Added_To* / *Assigned_To*) and real-time GPS data. No passenger entity required.
Favourite Routes (FR-029–FR-032)	Stored locally on the passenger's mobile device. Not persisted in the server-side database (BR-16).

Task Division

Table 16: Task Distribution for Milestone 2

Team Member	Role	Responsibilities
Ali Abbas (NUM-BSCS-2024-06)	Database Design	Identify entities and attributes, define all relationships and cardinalities, design ERD using Crow's Foot notation
Laiba Taj (NUM-BSCS-2024-31)	Documentation	Document attributes and constraints, format report in LaTeX, create final deliverable
Mehroz Ali Khan (NUM-BSCS-2024-34)	Validation	Verify cardinalities against SRS, define business rules, ensure normalization and requirement coverage, verify passenger data scope, cross-check ERD attribute and relationship names

Conclusion

This report presents the complete Entity-Relationship Diagram (ERD) for the Electric Bus Location Tracker System database, designed using **Crow's Foot notation**. The design covers all entities required by the SRS document and includes:

- **9 regular entities**: User, Driver, Admin, Bus, Route, Stop, Schedule, Duty_Assignment, and Report
- **1 associative entity**: Route_Stop_Detail (resolving the M:N Route–Stop relationship)
- **11 relationships**: All captured using Crow's Foot notation with ERD-matched names: *Is_A*, *Performs*, *Assigns*, *Assigned_To*, *has*, *Added_To*, *Included_In*, and *generates*
- **52 attributes**: Fully defined with appropriate data types, primary keys (Solid-underlined), foreign keys (Dotted-underline), and constraints
- **15 business rules**: Enforcing data integrity, scheduling constraints, system logic, and correct passenger data scoping

A critical design decision validated in this milestone is that **Passengers are not stored in the database** (SRS FR-005 restricts authentication to Driver and Admin roles only), and **favourite routes are stored locally** on the passenger's mobile device. This correctly scopes the relational database to authenticated users and their operational data.

The ERD follows Crow's Foot notation standards and the database has been normalized to 3NF, ensuring data consistency and eliminating redundancy. All entities and relationships have been mapped to the functional requirements in the SRS document, confirming complete requirement coverage. This database design serves as the foundation for the

MySQL implementation in the subsequent milestone.

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Appendix A: AI Tools Disclosure

Table 17: AI Tools Usage

Tool Used	Purpose	Date
ChatGPT (OpenAI)	Entity identification guidance and attribute definition assistance	March 2026
Claude (Anthropic)	ERD validation, cardinality verification, business rules formulation, passenger data scoping analysis, ERD–report cross-checking, and report review	March–May 2026

Prompts Used

1. **Prompt 1:** "Validate my supertype-subtype structure for User with Driver and Admin subtypes. What are the correct cardinalities in Crow's Foot notation?"
2. **Prompt 2:** "Generate appropriate business rules for a transportation database with driver duty assignments and bus scheduling."
3. **Prompt 3:** "Review my Milestone 2 report and identify any missing entities, relationships, or sections based on the ERD and rubric."
4. **Prompt 4:** "Verify my ERD against my report: are all relationship names, entity names, and attribute names consistent between the diagram and the written report?"

AI Usage Statement

AI tools were used solely for guidance, validation, and formatting assistance. All entity definitions, relationship identifications, and business rules were reviewed and approved by

the team based on the SRS requirements. Final decisions were made by team members.

Appendix B: Crow's Foot Notation Legend

Table 18: Crow's Foot Notation Symbols Reference

Symbol	**Meaning**
Rectangle (entity box)	Entity
Underlined attribute name	Primary Key
Italic attribute name	Foreign Key
Single vertical bar close to entity	Minimum/Maximum of 1 (mandatory, exactly one)
Circle (O) close to entity	Minimum of 0 (optional participation)
Crow's foot (3-pronged fork)	Maximum of N (many)
`||----|<O`	One-to-Many: mandatory one side, optional many side
`O|---O||`	One-to-One: optional both sides
Rectangle with PK + FK columns	Associative entity (resolves Many-to-Many)

End of Milestone 2 Report

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